

BrickAtlas: Supplementary Evidence

Development Manuscript
Author metadata intentionally omitted

Task	Model	RGB	Layers	Symbolic
reconstruct	GPT-4.1 mini	0/4	0/4	4/4
reconstruct	Gemini Flash	0/4	0/4	4/4
complete	GPT-4.1 mini	0/4	0/4	4/4
complete	Gemini Flash	0/4	0/4	4/4
repair	GPT-4.1 mini	0/4	1/4	3/4
repair	Gemini Flash	2/4	0/4	3/4

Table 1. Earlier composition-v1 pilot only. Gemini Flash denotes Gemini 2.5 Flash. These 112 API calls are not v2 model evaluation.

1. Earlier Neural Pilots

Two preceding releases provide neural pipeline evidence but not a v2 leaderboard. A 576-object composition pilot evaluated four held-out objects with GPT-4.1 mini and Gemini 2.5 Flash. Each model answered 56 cases, for 112 API requests. Table 1 shows visual task counts; all cells have only four objects. Both models failed ordinary and layer reconstruction/completion, but succeeded with the privileged symbolic reference. This suggests a useful perception-versus-structure diagnostic, not a statistically established model comparison. V2 also changes some scoring rules, including full-structure requirements under layer disclosure; cross-version scores must not be pooled.

A further Gemini control reuses the identical first answer and input for 12 ordinary-condition generation/reconstruction/repair cases. One additional call either requests reflection without new scene information or supplies target-blind legality feedback. Both arms remain at 2/12 exact successes, with no gained or lost successes. The 24 new requests are billed once; reused historical receipts are not charged again. Arms ran sequentially rather than interleaved, and the sample is too small to establish that validation tools are generally ineffective.

An older Grid-v1 text/structure pilot uses Qwen3-0.6B and rank-8 LoRA on query/value projections, with 1,146,880 trainable parameters, AdamW learning rate 2×10^{-4} , and 120 optimizer steps per trained run. The maximum sequence length is 3,072; overlength training examples are excluded rather than truncating answers. Greedy in-

Condition	Seed	Steps	Tokens	Success
base	17	0	0	0/8
single	17	120	195,292	2/8
multi	17	120	173,957	1/8
multi	29	120	176,765	1/8
multi	43	120	173,137	2/8

Table 2. Earlier Grid-v1 local text/structure pilot, eight test cases per row. Tokens count processed input plus output training tokens. Base/single have one seed each. This is not v2 or multimodal training.

ference allows 1,200 new tokens. Single-task training uses editing; multi-task training uses relations, generation, editing, and planning. Table 2 reports actual outputs. All successes are editing cases. Multi-task seeds average 16.7% with a sample standard deviation of 7.2 percentage points, without evidence of broad transfer.

The single/multi runs are step-matched, not token-matched, and use different supervision diversity. A lower validation NLL is not evidence of execution success. Four adapters, five prediction sets, and all 480 optimizer-step records are bundled. Training used an Apple M4 Pro with 48 GB memory and MPS; local computation consumed real resources despite having no API bill.

2. Evidence Boundaries

The six-work-package research contract separates positioning, data, measurement, model experiments, independent review, and delivery. Same-machine runtime-free replay is not independent reproduction. Review packets contain no completed human judgments until real reviewers submit version-matched labels; disagreement is not silently adjudicated. No human-designed dataset is treated as imported merely because a paper or connector annotation is publicly accessible. Permission, assets and redistribution rights are separate gates.

3. V2 Training And Verification

The local pilot uses SmolVLM-256M-Instruct revision 7e3e67ed, unquantized. Ninety-six training and 24

disjoint validation objects provide seven task families (672/168 examples) and actual PNG views. Language-side query/value rank-8 LoRA has 460,800 trainable parameters. Single-reconstruction, multi-task and leave-edit use seeds 17, 29 and 43. Every trained run uses 120 updates, exactly 512 supervised answer tokens per update, AdamW at 2×10^{-4} , and gradient clipping at 1. Full prompts and answers are retained; deterministic answer-token subsampling matches supervision, not input/image tokens, time or compute.

Validation NLL uses two fixed examples per task. Training uses PyTorch/MPS; formal inference uses MLX with a 2,200-token cap on the same 337 cases. Token, pixel and mask digests agree across preprocessing backends, but floating-point kernels need not produce identical outputs. All 840 training/validation prompt prefixes retain token identity when answers are appended; maximum complete length is 5,176 tokens. Compact test-oracle answers use at most 1,793 tokens. Oracle lengths are private diagnostics, never appended to model inputs. The completed matrix retains 3,370 predictions, 1,080 updates and 552,960 supervised tokens. Each of nine adapters has 60 adapted matrices and 411 unchanged tensors independently checked under the actual bfloat16 cast-before-add merge semantics.

The calibration pool contains 468 questions from 36 validation objects excluded from the original study and training/validation bundle. Selection cycles through available size bins within each of six policies; missing bins remain explicit. Portable inputs contain 720 image uses and 144 PNGs checked against public input, hashes and non-blank rendering. The earlier train/validation/test render audit checks 1,177 records, 1,970 image uses and 626 regenerated images on the same browser/GPU stack, not arbitrary rasterizers.

The casebank verifier checks 476,760 serialized rows across 14 shards, regenerates every ground truth and reconstructs baseline state chains. Browser checks cover desktop and mobile, nonblank scenes, camera interaction, input/truth isolation and export. Additional development responses are replayed offline and are not retroactively included in older browser-audit counts. Reproduction shares pinned Node dependencies; it is not an independently installed operating system or a second person’s training experiment.

4. Candidate Controls

For the initial pose probes, all candidate structures are legal, use identical part identities, yaw and height, and differ in horizontal position. Correct labels are balanced across the four selected scenes. Repeating a question and permuting its candidates are distinct interventions. The follow-up contract compares permutation mismatch to same-order repeat mismatch within each source/model, with temporal or

Development condition	Success	Abstain	Sources
Pose/choice-rgb	4/4	0	4
Pose/choice-permuted	4/4	0	4
Pose/choice-no-image	0/4	4	4
Order/original	12/12	0	12
Order/repeat	12/12	0	12
Order/permuted	12/12	0	12
Order/no-image	0/12	12	12

Table 3. Specialized known-renderer baseline. The 60 conditions reuse only 16 sources. No-image zeros are deliberate abstentions, not random-guess accuracy. This is not an additional neural model.

der balanced across source groups. Source-level bootstrap intervals describe only this selected development sample; they do not establish population coverage or stable model rankings.

4.1. Renderer-Aware Analysis by Synthesis

To test whether candidate locations contain usable image signal, a post-hoc deterministic baseline consumes only public BOM, candidate poses and RGB views. For the two-piece blue-on-gray domain, the supplied positive query height and support rules imply a grounded gray piece. Its horizontal positions and both distinct yaw footprints are enumerated under minimum-corner and contact constraints, not obtained from ground truth. Every legal candidate scene is rendered with the public renderer.

We threshold blue chroma ($B > 80$, $B > 1.25R$, $B > 1.1G$) inside image rows 32–447, excluding title/footer and neutral grid/text. A candidate receives the smallest mean mask-IoU distance over its legal hypotheses in top, front and side views. Equal candidate scores or missing blue masks cause abstention. No-image inputs abstain without rendering. The thresholds and enumeration are fixed in code before this run, but the method was designed after the preceding development experiments.

All 44 image-conditioned responses succeed, representing only 16 original sources with repeated/permuted conditions. All 16 permutations preserve the selected pose. Cyclic replacement with the next source’s views causes one abstention and changes 9/15 jointly answered selections; these mismatched BOM/image inputs test dependence on pixels, not valid-task accuracy. The implementation has no calibrated out-of-domain rejection threshold for nonblank inputs. The run generates 384 candidate views in 29.7 seconds on the local browser stack, including 60 predictions and 16 image replacements, with no API calls. Cached templates enable offline rescoring.

Using the exact renderer, canonical colors and candidate information strongly favors this method. It is an algorithmic simulator-aware control, not a general reconstruction system, novel learned model, independent human validation or

confirmation on new sources. Original VLM prompts also contained an isometric view, which this baseline does not use. Prediction production precedes loading evaluator targets for scoring.

5. Resource Accounting

The earlier 363-call campaign cost \$0.45102511, and the 730-call v2 study cost \$1.50635072. The four-model screen and both ladder versions add 228 calls and \$0.695073176; pose probes add 96 calls and \$0.162861212. The 192-call repeat/permutation study costs \$0.282100356. The full campaign totals 1,609 settled calls and \$3.097410574 under the original \$4.50 cap. No ledger is reset between studies. Local training and inference consume compute but do not incur API charges.